

Mid-term Project - Product Review Sentiment Analysis

System Prompt

As a Product Manager for a multinational manufacturer of computers, phones, laptops, and hardware, you are responsible for utilizing advanced AI tools to analyze customer reviews. Your goal is to extract critical information to enhance product quality and customer satisfaction. In this task, you will act as a Product Analyst, directly engaging with customer feedback to identify key insights and actionable recommendations. This involves structuring raw customer review data into clear, actionable formats to inform strategic product decisions.

Task Description:

Your primary objective is to process textual reviews from customers, extracting and structuring them into predefined categories to assess and enhance the customer experience. By doing so, you will help bridge the gap between customer feedback and actionable product development strategies. The insights you generate will directly influence product improvements and customer support enhancements. It is crucial that the responses you provide are detailed, thorough and comprehensive. The items to extract are:

"Review Date":

Extract the date from the beginning of the review text and save as <Review _Date>. Output <Review _Date> in the format [YYYY-MM-DD]. If the <Review _Date> is not explicitly mentioned then:

If a purchase date is provided: Save this as <Purchase _Date>.

If a duration (e.g., "5 years", "it's been 3 years", "4 months") is mentioned: Save this as <Duration>.

If <Duration> is provided: Calculate the <Calculated _Review _Date> as follows:

<Calculated _Review _Date> = <Purchase _Date> + <Duration>

For example:

<Purchase _Date> = 5/6/2018

<Duration> = 1 year

<Calculated _Review _Date> = 5/6/2018 + 1 year

<Calculated _Review _Date> = 5/6/2019

Output the review date as "Review Date: Not found, purchase date <Purchase_Date> duration <Duration> calculated review date <Calculated_Review_Date>".

If <Duration> is not provided: Output as "Review Date: Not found, purchase date <Purchase_Date>".

Reflect on the output for <Review Date> and ensure it explicitly follows the output instructions for <Review Date>.

"Product or service": Extract the product or service mentioned in the review.

"Review Rating": Extract the review rating. For example 1/5 or 4/5.

"Review Summary": Provide a summary of the review in 100 words or less. Focus on capturing the main points and overall sentiment.

"Improvement Areas": List any specific advice, suggestions or issues mentioned by the customer that could help improve the product or service. Prioritize actionable items.

"Competitors": If competitors' experiences with previous products are mentioned, explicitly output "Competitor: <Competitor> " followed by a summary of the customer's experience with the competitor. If multiple competitors are mentioned, separate them with commas. If no competitors are mentioned, output "None".

"Overall Sentiment": Based on the review content and Review Rating, classify the overall sentiment as Positive, Neutral, or Negative.

Each review will start with Review-X where X is a number. Ignore empty lines and in and between reviews. The responses you provide must be thorough and comprehensive. Do not output anything until I supply reviews.

Test Data

I used the LLM to fabricate three additional reviews (Review-3 => Review-5) in order to further test the robustness of my system prompt.

Review-1

[2023-08-24] - rating 2/5 - I bought the ZephyrBook Pro 2023 with high hopes, but it has been nothing but a disappointment. From the moment I powered it on, it felt like a terrible investment. The heating issue is absurd; this laptop heats up faster than a toaster. It's like having a portable oven on my lap. I can't even use it for more than 30 minutes without feeling like my thighs are on fire. The customer support experience has been equally horrendous. When I reached out to them about the heating problem, they brushed me off, saying it's normal for a laptop to get warm. Warm? No, this is scalding hot! It's like they expect me to carry a fire extinguisher with me while using it. Now, let me tell you about the competition. My friend owns the sleek and efficient StarLite 2023 laptop, and it's miles ahead. Not only does it stay cool as a cucumber, but it also runs like a dream. The ZephyrBook Pro 2023 can't hold a candle to it. My advice to the

company is simple: Invest in better hardware to manage the heating issue, and please, for the love of technology, improve your customer support. I regret not going for the StarLite 2023, and I wish I had made a wiser choice. Don't make the same mistake I did.

Review-2

rating 3/5 I can't express how delighted I am with my LuminaFlex X1 laptop. I purchased this in March 2020 and it's been 3 years. From the moment I opened the sleek packaging, I knew I had made a wise choice. This laptop has exceeded all my expectations, and I am thrilled to write this glowing review. The performance of the LuminaFlex X1 is top-notch. It handles all my tasks effortlessly, whether it's multitasking for work or enjoying some gaming during my downtime. The battery life is impressive, allowing me to work or play without constantly worrying about charging. But what truly sets LuminaFlex apart is their customer-oriented approach. The company understands the value of long-term customers. Not only do they offer a fantastic product, but they also provide extended warranty options and exclusive benefits for loyal customers. It's like they genuinely appreciate our trust in their brand. My advice to the company, if I may offer one, is to continue this customer-centric approach. Perhaps even consider extending the warranty period for all customers, as it provides peace of mind and further solidifies their commitment to excellence. In conclusion, if you're in the market for a laptop, look no further than the LuminaFlex X1. It's not just a laptop; it's an experience. Kudos to the LuminaFlex team for creating such a remarkable product and for valuing their customers like no other.

Review-3

[2023-07-15] - rating 4/5 - I purchased the QuantumTab S5 earlier this year, and it has been a solid performer. The display is bright and clear, perfect for watching movies and reading. However, the battery life could be better. It barely lasts a full day with moderate use. I also had a minor issue with the charging port, but customer support was quick to assist and resolve the problem. Overall, a good tablet with room for improvement.

Review-4

rating 1/5 - I bought my EchoPhone Z on December 1, 2022, and it has been nothing but trouble. The phone frequently crashes, and the camera quality is subpar. I contacted customer support multiple times, but they were unhelpful and dismissive. My previous phone, the ProVision Q, was much more reliable. I regret switching brands and would not recommend the EchoPhone Z to anyone.

Review-5

[2024-01-10] - rating 5/5 - NimbusLaptop V2 has been an excellent purchase. I've been using it for six months now, and it handles everything from video editing to gaming seamlessly. The battery life is impressive, and it charges quickly. I also appreciate the responsive customer service when I had questions about setup. Highly recommend!

Output using ChatGPT3.5Turbo in Azure OpenAI Studio

I used the ChatGPT3.5Turbo model in Azure OpenAI Studio with Chat playground.

Parameter Values

Max Response: 4000

Temperature: .75

Top P: .95

Frequency Penalty: 0

Presence Penalty: 0

Review-1

Review Date: 2023-08-24

Product or Service: ZephyrBook Pro 2023

Review Rating: 2/5

Review Summary: Heating issue is absurd, and customer support was dismissive.

Improvement Areas: Invest in better hardware to manage heating issues and improve customer support.

Competitors: Competitor: StarLite 2023, which stays cool and runs like a dream.

Overall Sentiment: Negative

Review-2

Review Date: Not found, purchase date March 2020 duration 3 years calculated review date 2023-03-01

Product or Service: LuminaFlex X1 laptop

Review Rating: 3/5

Review Summary: Delighted with the product and impressed with the customer-oriented approach.

Improvement Areas: Consider extending warranty period for all customers.

Competitors: None

Overall Sentiment: Positive

Review-3

Review Date: 2023-07-15

Product or Service: QuantumTab S5

Review Rating: 4/5

Review Summary: Solid performer with a bright and clear display.

Improvement Areas: Battery life could be better.

Competitors: None

Overall Sentiment: Neutral

Review-4

Review Date: Not found, purchase date December 1, 2022 duration Not found

Product or Service: EchoPhone Z

Review Rating: 1/5

Review Summary: Nothing but trouble with frequent crashes and subpar camera quality.

Improvement Areas: None mentioned.

Competitors: Competitor: ProVision Q, which was much more reliable.

Overall Sentiment: Negative

Review-5

Review Date: 2024-01-10

Product or Service: NimbusLaptop V2

Review Rating: 5/5

Review Summary: Handles everything seamlessly from video editing to gaming.

Improvement Areas: None mentioned.

Competitors: None

Overall Sentiment: Positive

Output using standard ChatGPT3.5Turbo

I used the ChatGPT3.5Turbo from <https://chatgpt.com/g/g-F00faAwkE-open-a-i-gpt-3-5> to be able to compare and contrast output using my prompt and data.

Review-1

Review Date: 2023-08-24

Product or Service: ZephyrBook Pro 2023

Review Rating: 2/5

Review Summary: The ZephyrBook Pro 2023 has severe heating issues, making it uncomfortable to use for more than 30 minutes. Customer support was dismissive about the problem. The reviewer regrets not purchasing the StarLite 2023, which they mention performs much better.

Improvement Areas:

- Address severe heating issues.
- Improve customer support responsiveness and quality.

Competitors: Competitor: StarLite 2023 - Customer reported it stays cool and runs efficiently, far outperforming the ZephyrBook Pro 2023.

Overall Sentiment: Negative

Review-2

Review Date: Not found, purchase date March 2020, duration 3 years, calculated review date 2023-03-01

Product or Service: LuminaFlex X1

Review Rating: 3/5

Review Summary: The LuminaFlex X1 has met all the reviewer's expectations with excellent performance and battery life. The reviewer appreciates the company's customer-centric approach and extended warranty options. They suggest continuing this approach and considering extending the warranty period for all customers.

Improvement Areas:

- Consider extending the warranty period for all customers.

Competitors: None

Overall Sentiment: Positive

Review-3

Review Date: 2023-07-15

Product or Service: QuantumTab S5

Review Rating: 4/5

Review Summary: The QuantumTab S5 has a bright and clear display, making it great for media consumption. However, its battery life is subpar, and there was a minor issue with the charging port that customer support resolved quickly.

Improvement Areas:

- Improve battery life.

Competitors: None

Overall Sentiment: Positive

Review-4

Review Date: Not found, purchase date December 1, 2022

Product or Service: EchoPhone Z

Review Rating: 1/5

Review Summary: The EchoPhone Z has been problematic with frequent crashes and poor camera quality. Customer support has been unhelpful. The reviewer regrets switching from their previous phone, the ProVision Q, which was more reliable.

Improvement Areas:

- Fix frequent crashing issues.
- Improve camera quality.
- Enhance customer support.

Competitors: Competitor: ProVision Q - Customer found it more reliable than the EchoPhone Z.

Overall Sentiment: Negative

Review-5

Review Date: 2024-01-10

Product or Service: NimbusLaptop V2

Review Rating: 5/5

Review Summary: The NimbusLaptop V2 has been an excellent purchase, performing well for video editing and gaming. It has impressive battery life and charges quickly. Customer service was responsive and helpful during setup.

Improvement Areas: None mentioned.

Competitors: None

Overall Sentiment: Positive